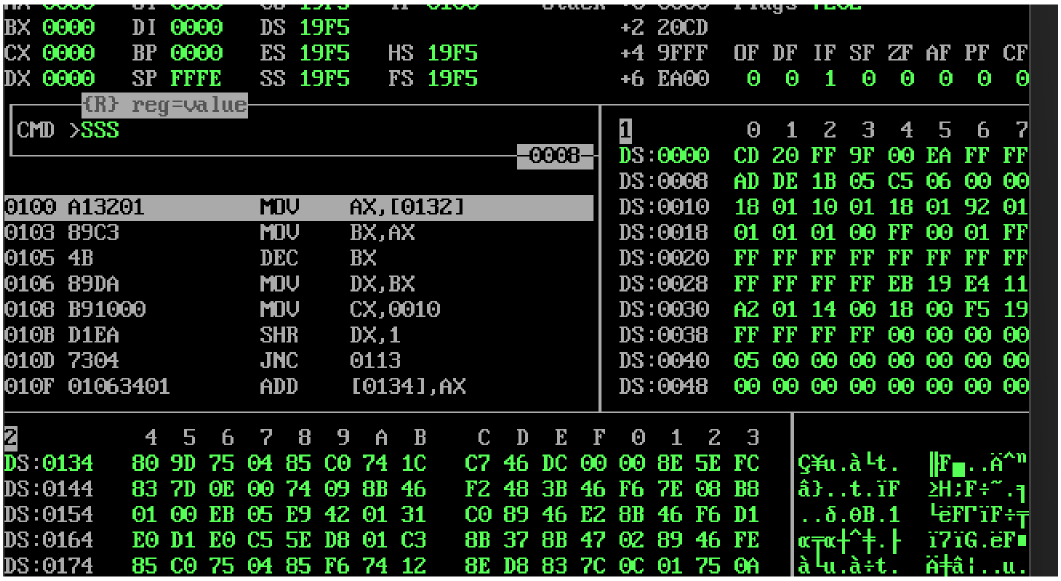
MUHAMMAD TAHA
22F-3277
BCS-3C
LAB 7

Q3. Write a Subroutine that calculates the factorial of a given number using following requirements.

- **Number is 10**
- **Taking all parameter from the stack**
- **Saving and restoring register values**
- **All stack must be cleared before ending the program**
- **Mnemonic "mul" is not allow to use.**

```
org 0x0100

jmp start

hi:
        push bp
        mov bp,sp
        sub sp,2

        mov bx, [bp+4]

factorial:
  dec bx
  mov dx, bx
  mov cx, 8

l1:
  shr dx, 1
  jnc skip

  add [bp-2], ax

skip:
  shl ax, 1
  dec cx
  jnz l1

  cmp bx, 1
  je done
  mov ax, [bp-2]
  mov word [bp-2], 0

done:
```

```
    cmp bx, 1
    jne factorial

            pop bp
            mov sp,bp
            pop bp
            ret

start:
            mov ax , num
            push ax
            call hi


mov ax, 0x4c00
int 0x21

num: db 8h
result: db 0
```

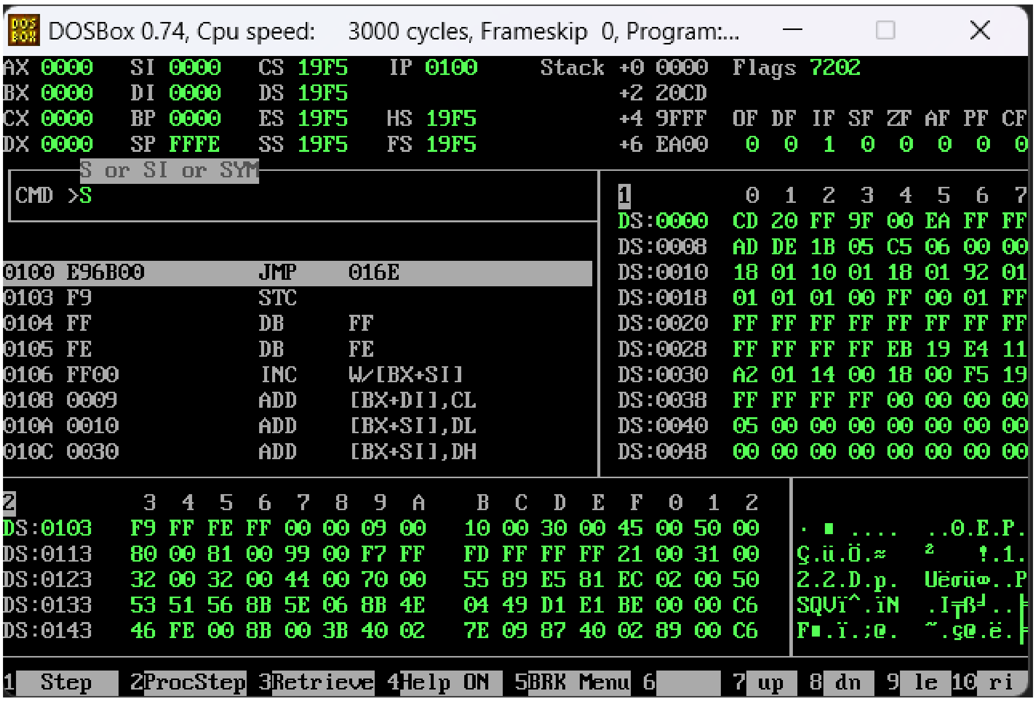


Q4. Write assembly code to implement bubble sort with mentioned following requirements.

- **Use subroutine for calling**
- **Use a local variable(stack) to store the swap flag.**
- **Taking all parameter from the stack**
- **Saving and restoring register values**
- **All stack must be cleared before ending the program**
- **array1: dw 80, -7, -2, 45, 81, 50, 10, 9, 30, 99, 0**
- **array2: dw 70, 32, 21, -3 , -1, 31, 44, -9, 32**

```
[org 0x0100]
jmp start

array1: dw 80h,-7h,-2h,45h,81h,50h,10h,9h,30h,99h,0h
array2: dw 70h,32h,21h,-3h,-1h,31h,44h,-9h,32h

bubblesort:
   push bp
   mov bp, sp
   sub sp, 2
   push ax
   push bx
   push cx
   push si

   mov bx, [bp+6]
   mov cx, [bp+4]
   dec cx
   shl cx, 1

mainloop:
   mov si, 0
   mov byte [bp-2], 0

innerloop:
   mov ax, [bx+si]
   cmp ax, [bx+si+2]
   jle noswap

   xchg ax, [bx+si+2]
   mov [bx+si], ax
   mov byte [bp-2], 1

noswap:
   add si, 2
   cmp si, cx
   jne innerloop

   cmp byte [bp-2], 1
   je mainloop

   pop si
   pop cx
```

```
    pop bx
    pop ax
    mov sp, bp
    pop bp
    ret 4

start:
    mov ax, array1
    push ax
    mov ax, 11
    push ax
    call bubblesort

    mov ax, array2
    push ax
    mov ax, 9
    push ax
    call bubblesort

    mov ax, 0x4c00
    int 0x21
```



Array1 starts from 0103
Array2 starts from 0119